

# Christian Döring

Graduate Student

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## Education

2023 – 2026  
April March

**M.Sc. Electrical and Computer Engineering,**  
*Technical University of Munich*

Thesis Title: Representing Mesoscale Detail in Neural Materials

2019 – 2023  
October March

**B.Sc. Electrical and Computer Engineering,**  
*Technical University of Munich*

Thesis Title: Evaluation of Differentiable Inverse Rendering using Multi-View RGB Data

2011 – 2019  
September June

**Abitur (A-Levels),** *Gymnasium Bruckmühl*

## Publications

2026

**Real-time Rendering with a Neural Irradiance Volume**

Arno Coomans, Giacomo Nazzaro, Edoardo A. Dominici, Christian Döring, Floor Verhoeven, Konstantinos Vardis, Markus Steinberger

📄 arXiv

2024

**Real-time Neural Rendering of Dynamic Light Fields**

Arno Coomans, Edoardo A. Dominici, Christian Döring, Joerg H. Mueller, Jozef Hladky, Markus Steinberger

📄 Project

📄 In Computer Graphics Forum (EG), 2024

## Experience

2025 – 2026  
May May

**Intern, Realistic Graphics Laboratory, EPFL**

○ Representing Mesoscale Detail in Neural Materials

2025 – 2026  
September March

**Masters Thesis, Realistic Graphics Laboratory, EPFL**

○ Representing Mesoscale Detail in Neural Materials

2025 – 2025  
April August

**Research Intern, NVIDIA Zurich**

○ Function Level Caching for Dr.Jit and Mitsuba 3

○ Hash Grid for Dr.Jit

○ Differentiable Radio Frequency Modeling with Sionna RT

2024 – 2025  
April April

**Research Working Student, Huawei Technologies Munich**

○ Development on Dr.Jit/Mitsuba 3

○ Real-time Neural Rendering Research

2023 – 2024  
August February

**Research Intern, Huawei Technologies Munich**

- Real-time Neural Rendering Research

2021 – 2021  
July August

**Embedded Systems Intern, Aurum GmbH Munich**

- Developed NFC library for STM32 in C

2017  
July

**Intern, Lauterbach GmbH**

2017  
July

**Intern, Electronic Theater Controls (ETC) Holzkirchen**

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## Side Projects

**Hephaestus**, Just In Time Compiler (JIT) for Vulkan, inspired by Dr.Jit. Implemented with own render graph solution. Includes cooperative matrix multiplication (KHR) and a port of tiny-cuda-nn in GLSL.

◆ Source

**Vulkan Path Tracer**, Path tracer written in Rust using the screen-13 library. It supports the Disney BSDF with Next Event Estimation.

◆ Source

**Mitsuba 3 Experiments**, Implementation of forward and differentiable path tracing algorithms in Mitsuba 3, such as ReSTIR GI and Large Steps in Inverse Rendering

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## Skills

Programming:

- **C/C++**, Rust
- **CUDA**, Vulkan
- **Python**, Lua
- **LaTeX**, Typst

Languages:

- **German** (native)
- English (B2+/C1)